

These frameworks define the appearance of an app. These also provide the basic app infrastructure and support for key technologies such as multitasking, touch based input, push notifications and many high-level system services.

High-level features available in Cocoa Touch layer are App extensions, Handoff, AirDrop, TextKit, UIKit Dynamics, Multitasking, Auto Layout, User Interface (UI) State Preservation, Apple Push Notification Service, Local Notifications and Gesture Recognisers, among others.

Described next are the frameworks of Cocoa Touch layer and the services these offer:

Address Book UI framework. Address Book UI framework (AddressBookUI.framework) is an Objective-C programming interface that you use to display standard system interfaces for creating new contacts and for editing and selecting existing contacts. This framework simplifies the work needed to display contact information in the app and also makes sure that the app uses the same interfaces as other apps, thus ensuring consistency across the platform.

UIKit UI framework. UIKit UI framework (UIKitUI.framework) provides view controllers for presenting standard system interfaces for viewing and editing calendar-related events.

GameKit framework. GameKit framework (GameKit.framework) implements support for Game Centre, which lets users share their game-related information online.

iAd framework. iAd framework (iAd.framework) lets you deliver banner based advertisements from your app. Advertisements are incorporated into standard views that you integrate into your UI and present when required.

MapKit framework. MapKit framework (MapKit.framework) provides a scrollable map that you can incorporate into your app's UI. Beyond just displaying a map, you can use the framework interfaces to customise

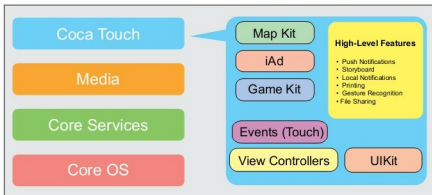


Fig. 4: Cocoa Touch layer

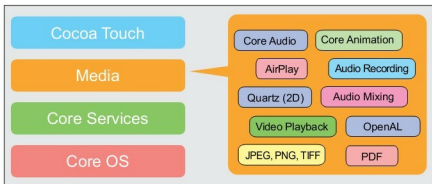


Fig. 5: Media layer

the map's content and appearance.

In addition to displaying maps, MapKit framework integrates with Maps app and Apple's map servers to facilitate directions.

Message UI framework. Message UI framework (MessageUI.framework) provides support for composing emails or SMS messages from your app. The composition support consists of a view-controller interface that you present in your app.

Notification Centre framework. Notification Centre framework (NotificationCenter.framework) provides support for creating widgets that display information in Notification Centre.

PushKit framework. PushKit framework (PushKit.framework) provides registration support for voice-over Internet protocol (VoIP) apps. This framework replaces the previous application program interfaces (APIs) for registering VoIP apps. Instead of keeping a persistent connection open, and thus draining the device's battery, an app can use this framework

to receive push notifications when there is an incoming call.

Twitter framework. Twitter framework (Twitter.framework) has been replaced by Social framework, which supports a UI for generating tweets and support for creating URLs to access Twitter service.

UIKit framework. UIKit framework (UIKit.framework) provides crucial infrastructure for implementing graphical, event-driven apps in iOS.

Media layer. Media layer contains graphics, audio and video technologies you use to implement multimedia experiences in your apps. Technologies in this layer make it easy for you to build apps that look and sound great.

iOS graphics technologies offer a wide range of support, working seamlessly with UIKit view architecture to make it easy to deliver content.

High-level features available in Media layer are UIKit graphics, Core Graphics framework, Core Animation, Core Image, OpenGL ES and